

Pharmaceutical Analytical Chemistry II

PCC/PCD 105

Week 5 lecture

OXIDATION REDUCTION TITRATIONS

Standard Oxidizing Agents

Potassium permanganate, KMnO_4

Cerium (IV), Ce^{4+}

Potassium iodate, KIO_3

Periodic acid, HIO_4 , H_5IO_6

Potassium bromate, KBrO_3

Potassium dichromate, $\text{K}_2\text{Cr}_2\text{O}_7$

Iodine, I_2

Standard Reducing Agents

Sodium thiosulfate, $\text{Na}_2\text{S}_2\text{O}_3$

Fe^{2+}

Standard Oxidizing Agents

Standard Oxidizing Agents	Reduced species	$E^0 / 1 \text{ M H}_2\text{SO}_4$
H_5IO_6	IO_3^-	+1.6
KMnO_4	Mn^{2+}	+1.51
Ce^{4+}	Ce^{3+}	+1.44
KBrO_3	Br^-	+1.44
$\text{K}_2\text{Cr}_2\text{O}_7$	Cr^{3+}	+1.33
I_2	I^-	+0.536

1) Potassium permanganate, KMnO_4

- * purple $\rightarrow$ self indicator
- * $\uparrow$ +ve $E^\circ \rightarrow$ strong oxidant
- * 2^{ry} standard....unstable....why?????



Brown

KMnO_4 oxidizes water (solvent) & this is enhanced by:

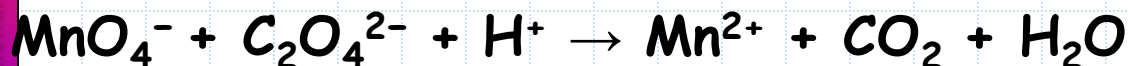
1-light...so stored in amber glass (dark colored)

2- H^+ (acidity)

3- Δ (heat)

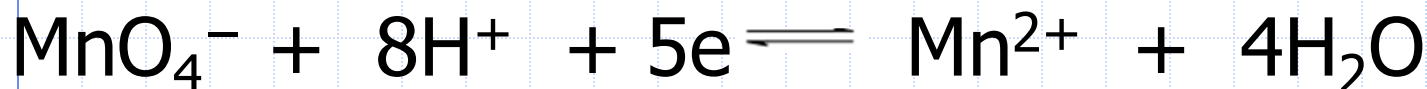
4- MnO_2autocatalytic degradation =autocatalyst

*standardized # oxalic acid or Na oxalate "1^{ry}"



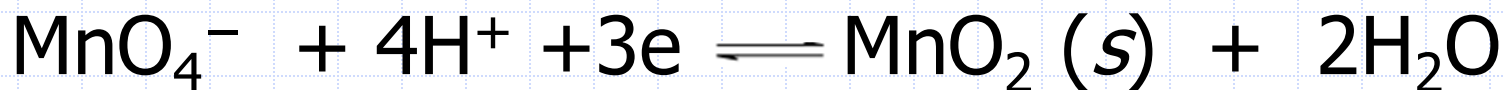
Reduction half reactions depend on **pH**:

◆ Strongly acidic solutions (pH ≤ 1)

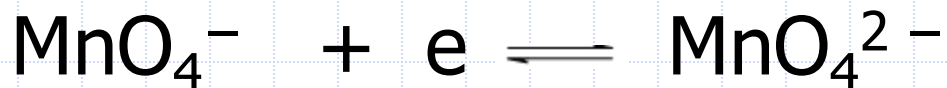


$$\text{eq. Wt} = \text{M.wt} / 5$$

◆ Neutral solutions



◆ Strongly alkaline solutions (2 M NaOH)

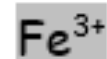
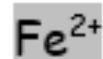
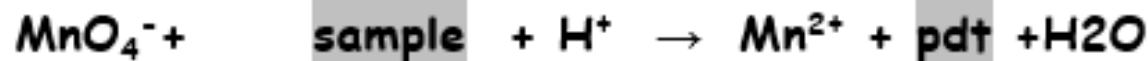


Applications

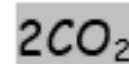
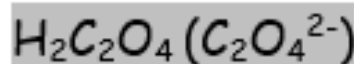
strong acidic Medium, 0.1 to 1 M sulphuric acid

Color change: colorless - pink

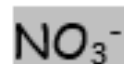
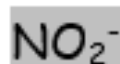
1) direct:



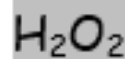
$$n_{\text{sample}} = 1$$



$$n_{\text{sample}} = 2$$



$$n_{\text{sample}} = 2$$



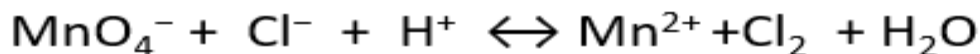
$$n_{\text{sample}} = 2$$

Interference due to chloride

For $\text{MnO}_4^- / \text{Mn}^{2+}$ $E^0 = +1.51 \text{ V}$

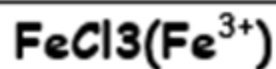
For $\text{Cl}_2 / \text{Cl}^-$ $E^0 = +1.36 \text{ V}$

- $\text{MnO}_4^- / \text{Mn}^{2+}$ system $>$ $\text{Cl}_2 / \text{Cl}^-$ system in E^0
- So MnO_4^- oxidizes Cl^- impurity (not a sample)

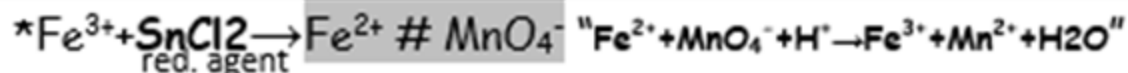


- Results in +ve error.
- Treatment: ZR reagent Will be discussed later

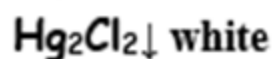
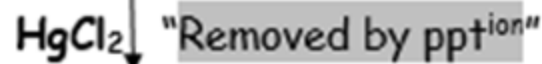
2) indirect:



Metal ions that form insoluble oxalate (e.g. Ca^{2+})



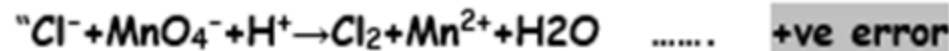
N.B. Xss SnCl₂ "Cl⁻" reacts with MnO_4^- **+ve error**



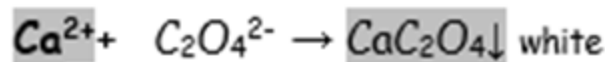
*Problems: presence of Cl⁻:

→ With Fe^{3+} .. **Yellow** color interfere with pink color at e.p.

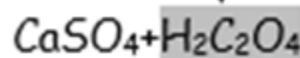
→ Cl^- is **reducing agent**...react with MnO_4^-



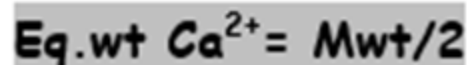
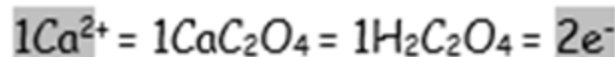
*soln. Zimmermann Reinhardt (ZR) reagent



1) filtration 2) $\text{H}_2\text{SO}_4 (\text{H}^+)$



, where $n=2$



Zimmermann- Reinhardt reagent

ZR Reagent: $\text{MnSO}_4 + \text{H}_2\text{SO}_4 + \text{H}_3\text{PO}_4$

a) MnSO_4 :

E° of $\text{MnO}_4^-/\text{Mn}^{2+} > \text{Cl}_2/\text{Cl}^- > \text{Fe}^{3+}/\text{Fe}^{2+}$ So MnO_4^- can oxidize both Fe^{2+} & Cl^- +ve error
, but by adding of MnSO_4



$$0.059 \quad [\text{Mn}^{2+}] \uparrow$$

$$E \downarrow = E^\circ - \frac{0.059}{5} \log \frac{[\text{Mn}^{2+}]}{[\text{MnO}_4^-] [\text{H}^+]^8}$$

, SO E of $\text{Cl}_2/\text{Cl}^- > \text{MnO}_4^-/\text{Mn}^{2+} > \text{Fe}^{3+}/\text{Fe}^{2+}$

b) H_3PO_4 : + $\text{Fe}^{3+} \rightarrow [\text{FeHPO}_4]^+$ cpx...

$$0.059 \quad [\text{Fe}^{2+}]$$

$$E \downarrow = E^\circ - \frac{0.059}{1} \log \frac{[\text{Fe}^{2+}]}{[\text{Fe}^{3+}] \downarrow}$$

, so * E of $\text{Fe}^{3+}/\text{Fe}^{2+} \downarrow$...more feasible Rx (i.e. facilitate it)

- Eliminate yellow color of FeCl_3

c) H_2SO_4 : source of acidity (make MnO_4^- potential suitable to act as oxidizing agent)

Determination of Fe^{2+} / Fe^{3+} mixture

- ◆ **(a) For Fe^{2+} content:** direct titration against standard MnO_4^-
- ◆ **(b) For total ($\text{Fe}^{2+} + \text{Fe}^{3+}$) content:** as in Fe^{3+} alone
- ◆ **(c) For Fe^{3+} content :** by difference

Determination of Oxalic acid-Sodium Oxalate Mixture

- **(a) For oxalic acid content:** titration against NaOH using ph.ph. indicator.
- **(b) For total content:** titration against standard MnO_4^- in acid medium.
- **(c) For sodium oxalate content:** by difference

Strongly alkaline solution

By back titration

◆ For org. Cpds e.g. Glycerol **Very slow reaction**



Glycerol + NaOH + known xss std. $\text{MnO}_4^- \rightarrow$ unconsumed MnO_4^-
back titrated # std. oxalic acid

Color change: pink -colorless

The produced MnO_4^{2-} "manganate" is green color which interferes with color change at e.p.

Barium (Ba^{2+}) is added to ppt manganate.



Eq. Wt of glycerol = M.wt/ 14